HARDWARE ASSEMBLY & REFERENCE SPECIFICATION

Earthquake & Tremor Alarm System

IoT Early Warning Seismic Detector & Acoustic Notification Module

System Scope: This hardware manual details the physical assembly of a localized, high-sensitivity seismic and structural vibration monitor. Built around an ESP32-WROOM-32 microcontroller, the system monitors ground tremors via an SW-420 digital vibration sensor. Acoustic alert signaling is executed using a 3V passive buzzer driven by a BC547 NPN transistor switch circuit to ensure complete current isolation and pin protection for the microcontroller.

BILL OF MATERIALS (BOM)

1. ESP32-WROOM-32 (38-Pin CP2102)
Main system processor, handles real-time digital edge detection and wireless alert communication.

2. SW-420 Vibration Sensor Module
Piezo-electric / roller ball vibration switch featuring onboard LM393 comparator for digital output.

3. 9042 Passive Buzzer (16Ω 3V)
Compact acoustic actuator requiring PWM drive signal from ESP32 to produce alarm frequencies.

4. BC547 NPN Transistor & 1kΩ Resistor
General-purpose NPN transistor driver paired with 0.25W base current limiting protection resistor.

5. MB-102 Solderless Breadboard
Standard 830-point breadboard with split power rails for modular circuit organization.

6. Male-to-Female Jumper Wires
20cm flexible interconnects for coupling header pins to breadboard points.

HARDWARE MASTER WIRING MATRIX

SOURCE COMPONENT	COMPONENT PIN	TARGET CONNECTION	ELECTRICAL / FUNCTIONAL NOTES
ESP32 Board	GND	**Breadboard GND Rail (-)**	Establishes common system ground reference.
ESP32 Board	3V3	**Breadboard Power Rail (+)**	Supplies clean 3.3V power to onboard peripherals.
SW-420 Sensor	VCC	**Breadboard Power Rail (+)**	Module operational voltage feed (3.3V nominal).
SW-420 Sensor	GND	**Breadboard GND Rail (-)**	Sensor ground return path.
SW-420 Sensor	DO (Digital Out)	**ESP32 GPIO 18**	Transmits logic HIGH/LOW vibration status state.
BC547 Transistor	Collector (Pin 1)	Buzzer Negative (-) Pin	Low-side switching node for acoustic load.
BC547 Transistor	Base (Pin 2)	**ESP32 GPIO 23** via 1kΩ	Switch control line; 1kΩ caps base current to ~2.6mA.
BC547 Transistor	Emitter (Pin 3)	**Breadboard GND Rail (-)**	Drains load switched current safely to ground rail.

STEP-BY-STEP HARDWARE ASSEMBLY

1 Establishment of Power Rails & Microcontroller Mounting

- **Mount the ESP32:** Place the ESP32-WROOM-32 module onto the breadboard, straddling the central divider ditch. Ensure all 38 header pins align cleanly with separate terminal rows.
- **Connect Common Rails:** Run a jumper wire from the ESP32 `3V3` pin to the red positive (+) outer rail. Connect a second jumper from an ESP32 `GND` pin to the black negative (-) outer rail.
- **Position the BC547 Transistor:** Face the flat side of the TO-92 BC547 housing toward you. Insert the three legs into three distinct adjacent breadboard terminal rows. From left to right, identify the pins as **Collector (C)**, **Base (B)**, and **Emitter (E)**.

2 SW-420 Sensor Integration

- **Mount Sensor Module:** Seat the SW-420 vibration sensor in an open breadboard area.
- **Supply Power:** Connect SW-420 **VCC** pin to the red positive rail (+), and **GND** to the black ground rail (-).
- **Route Signal Line:** Connect the **DO (Digital Output)** pin of the sensor directly to row **GPIO 18** on the ESP32 using a male-female jumper wire.

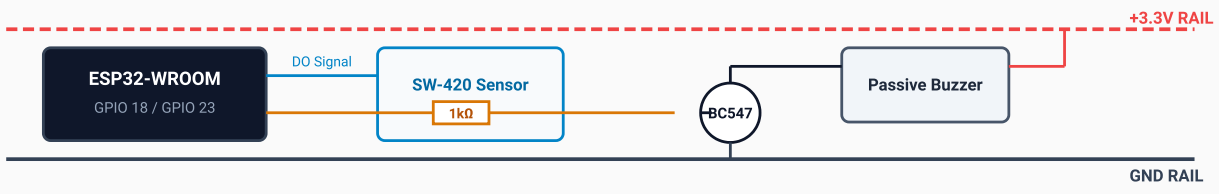
3 Driver Circuitry & Acoustic Buzzer Wiring

- **Insert Protection Resistor:** Span the **1kΩ resistor** between the breadboard row for **GPIO 23** and the center leg (Base) of the BC547 transistor.
- **Ground Transistor Switch:** Run a jumper wire from the right leg (Emitter) of the BC547 directly to the common ground rail (-).
- **Connect Passive Buzzer:** Connect the positive (+) long pin of the 9042 Passive Buzzer to the positive rail (+). Connect the negative (-) short pin directly to the row matching the left leg (Collector) of the BC547.

Critical Safety Check: Never connect the passive buzzer directly to the ESP32 GPIO pin. A 16Ω buzzer draws ~200mA at 3.3V, exceeding the maximum safe GPIO source limit of 12mA–40mA. The BC547 buffer transistor prevents GPIO destruction.

CIRCUIT TOPOLOGY SCHEMATIC REPRESENTATION

SYSTEM INTERCONNECT & LOW-SIDE DRIVER SCHEMATIC



EXPANDED REMOTE NOTIFICATION OPTIONS

Following hardware assembly, the ESP32 can be integrated with various internet and network messaging topologies for remote tremor notification. The available communication routes are organized below by display mode:

Telegram Bot API (Recommended Instant App-Free Route)

Workflow: The ESP32 sends direct HTTPS requests to the Telegram Bot API when digital vibration limits are exceeded.
Access Method: Accessible via any web browser using Telegram Web without requiring dedicated software. Provides push-style vibration alerts with minimal transmission latency.

WhatsApp Web / CallMeBot API Gateway

Workflow: The ESP32 triggers a light HTTP GET call to the public CallMeBot service endpoint.
Access Method: Messages are converted and delivered straight into your primary WhatsApp chat thread, accessible on desktop browsers via WhatsApp Web.

Direct-to-SMS Cellular Gateway (Twilio REST API)

Workflow: The ESP32 executes an authenticated REST API call to Twilio over Wi-Fi.
Access Method: Converts cloud packets into standard cellular SMS text messages sent directly to mobile phone numbers, operating independently of mobile data apps.

Local Network Emergency Dashboard (Onboard Web Server)

Workflow: The ESP32 hosts an embedded HTTP server providing real-time WebSocket state updates.
Access Method: Navigating to the ESP32 IP address on a local browser displays a visual status board that flips to a full-screen red warning with audio alerts upon sensor trigger.

Automated Email Dispatch (Encrypted SMTP Relay)

Workflow: The ESP32 connects directly to an SMTP email server using TLS encryption.
Access Method: Delivers high-priority alert emails containing timestamps and sensor trip counts directly to standard mail clients.